

Announcements 3/30

Unit 3 starts!

- Exam 3 at 11 AM on Fri 4/17 (week 11)

Assignments that are due soon

- Pre-Class Assignment Week 9-W

Office Hour

- Dr. Wang's: Tue 10-11 AM at KC 230
- Paige Kwan's: Mon 11:30 AM -12:30 PM at the TLC (Demille 160P)
- Saige Douglass's: Tue 12:00-1:00 PM at the TLC (Demille 160P)



Chapter 15: Chemical Kinetics



Dr. Wang

Learning Objectives

- Understand and interpret concentration vs. time graphs for simple chemical reactions.
- Write the mathematical expressions for rate of reaction for each reactant and each product given a chemical reaction
- Determine rates of reaction relating rates based on stoichiometric coefficients.
- Determine instantaneous and initial rates of reaction given concentration vs. time graphs for a reactant or product in the chemical reaction.
- Determine the rate law expression for a reaction given experimental reactant concentration and reaction rate data.
- Use the rate law expression to quantify the instantaneous reaction rate with specific reactant concentrations.

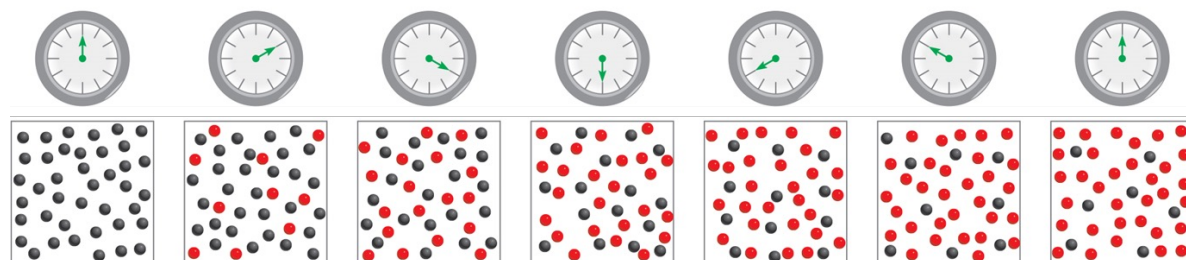
Chemical Kinetics

- Chemical kinetics studies the rate (or speed) of a reaction
- Thermodynamics predicts whether a reaction is able to happen spontaneously; Kinetics describes how fast or slow the reaction is.
- Reaction rate (r) = the change in concentration of a reactant/product over time (Δt)

$$\text{A} \rightarrow \text{B}$$

$$r = \frac{-\Delta [\text{A}]}{\Delta t} = \frac{\Delta [\text{B}]}{\Delta t}$$

(-) [Reactant] (+) [Product]

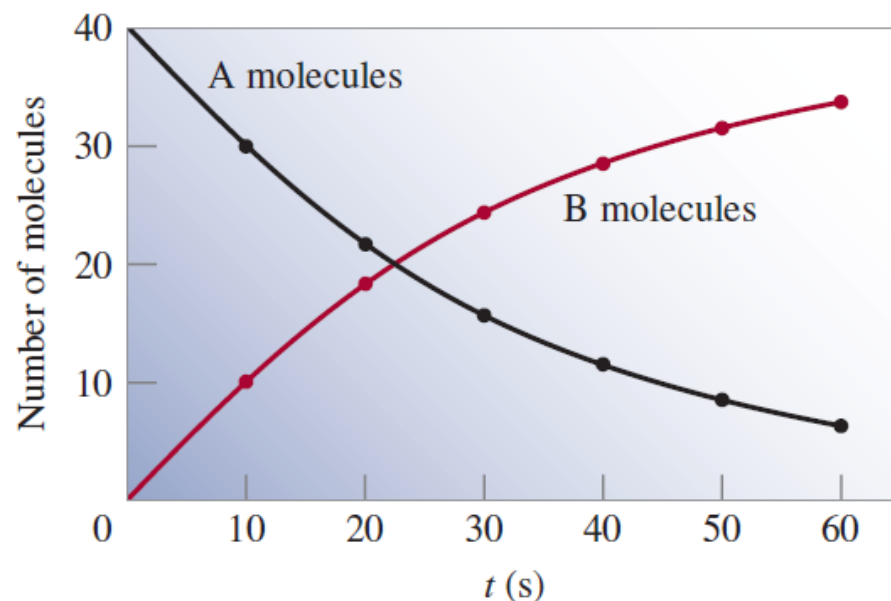


- Unit:

$$\frac{\frac{\text{mol}}{\text{L}}}{\text{s}}$$

or

$$\frac{\text{M}}{\text{s}}$$



Chemical Kinetics

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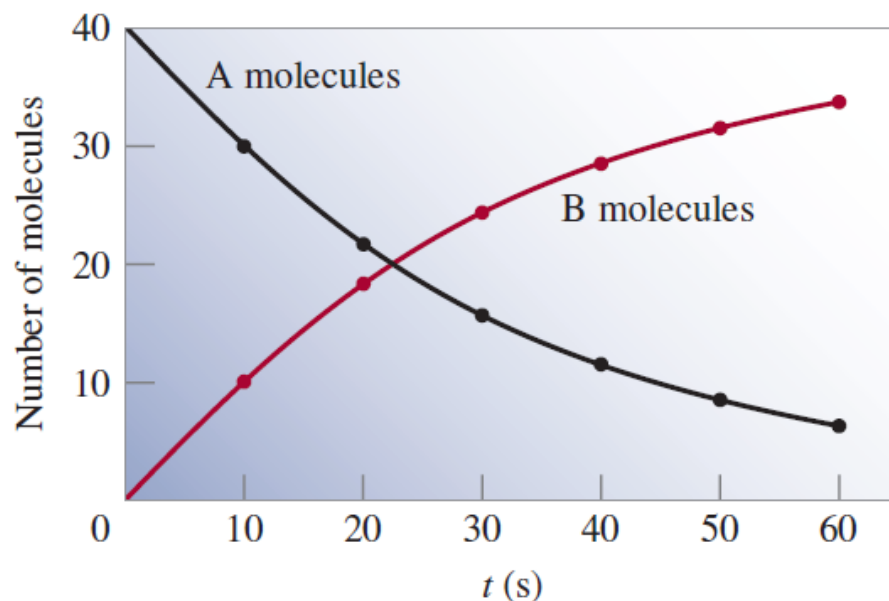
$$r = \frac{-\Delta[\text{A}]}{\Delta t} = \frac{\Delta[\text{B}]}{\Delta t}$$

- [Reactant] + [Product]

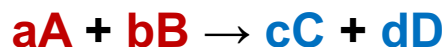
$$\text{A} \rightarrow \text{B}$$

$$r = \frac{\Delta}{\Delta t} = \frac{\Delta}{\Delta t}$$

- Unit:
 $\frac{\text{mol}}{\text{L}} / \text{s}$
 or
 M / s



Chemical Kinetics



- ➔ Rate of change in the concentration of a reactant/product:

$$\frac{\Delta[A]}{\Delta t} ; \frac{\Delta[B]}{\Delta t} ; \frac{\Delta[C]}{\Delta t} ; \frac{\Delta[D]}{\Delta t}$$

- ➔ Reaction rate (r) = the change in concentration of a reactant/product over time (Δt)

$$r = \frac{1}{a} \frac{-\Delta[A]}{\Delta t} = \frac{1}{b} \frac{-\Delta[B]}{\Delta t} = \frac{1}{c} \frac{\Delta[C]}{\Delta t} = \frac{1}{d} \frac{\Delta[D]}{\Delta t}$$

(-) [Reactant]
(+) [Product]

Which expression of reaction rate is correct for the reaction:
 $2H_2O_2(aq) \rightarrow 2H_2O(l) + O_2(g)$?

↙ rate of change in H_2O_2

A. $r = \frac{1}{1} \frac{\Delta[H_2O_2]}{\Delta t}$

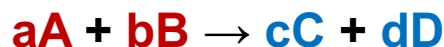
C. $r = \frac{1}{2} \frac{\Delta[H_2O_2]}{\Delta t}$

B. $r = \frac{1}{1} \frac{-\Delta[H_2O_2]}{\Delta t}$

D $r = \frac{1}{2} \frac{-\Delta[H_2O_2]}{\Delta t}$



Chemical Kinetics



- Rate of change in the concentration of a **reactant**/**product**:

$$A: \frac{\Delta A}{\Delta t} \quad B: \frac{\Delta B}{\Delta t} \quad C: \frac{\Delta C}{\Delta t} \quad D: \frac{\Delta D}{\Delta t}$$

- Reaction rate (r) = the change in concentration of a **reactant**/**product** over time (Δt)

↑
(+)
per equation

$$r = \frac{1}{a} \frac{-\Delta[A]}{\Delta t} = \frac{1}{b} \frac{-\Delta[B]}{\Delta t} = \frac{1}{c} \frac{+\Delta[C]}{\Delta t} = \frac{1}{d} \frac{+\Delta[D]}{\Delta t}$$

(-) [Reactant] (+) [Product]

Which expression of reaction rate is correct for the reaction:
 $2H_2O_2(aq) \rightarrow 2H_2O(l) + O_2(g)$?

A. $r = \frac{1}{1} \frac{\Delta[H_2O_2]}{\Delta t}$

C. $r = \frac{1}{2} \frac{\Delta[H_2O_2]}{\Delta t}$

B. $r = \frac{1}{1} \frac{-\Delta[H_2O_2]}{\Delta t}$

D. $r = \frac{1}{2} \frac{-\Delta[H_2O_2]}{\Delta t}$



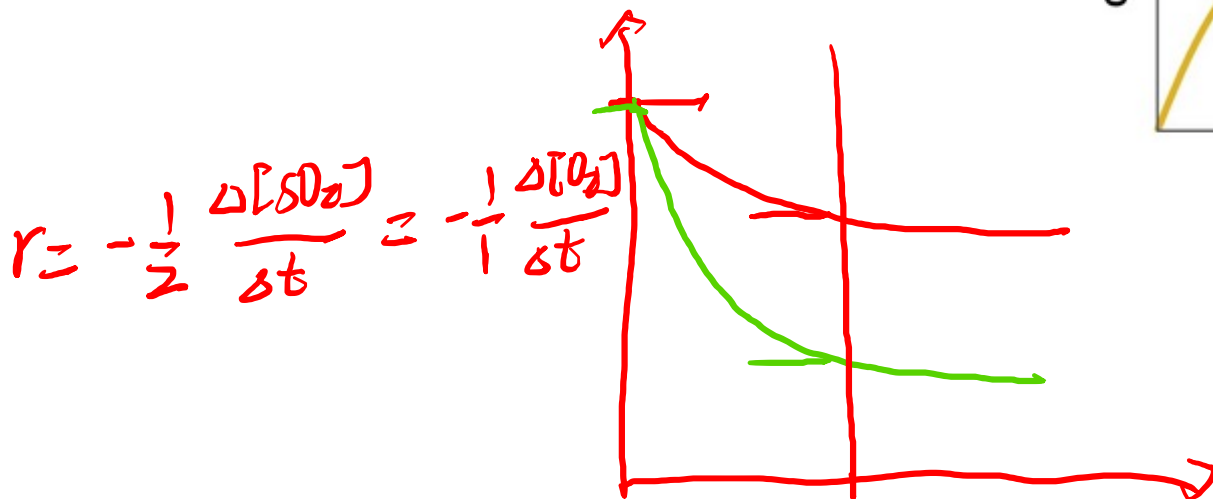
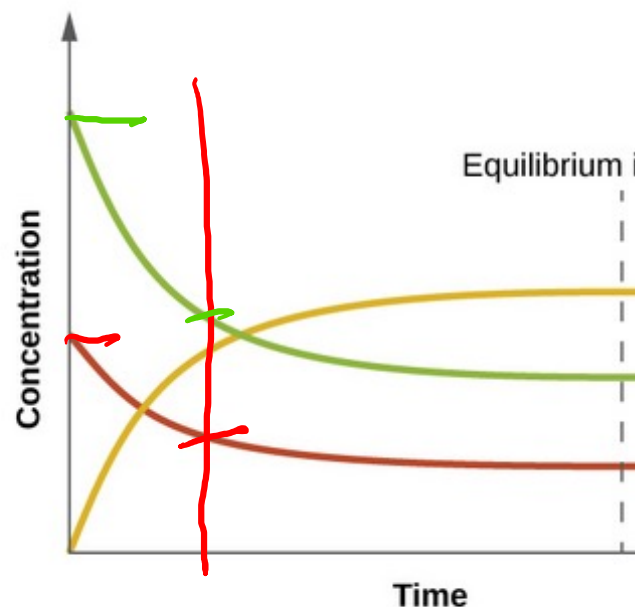
Your Turn!

The following graph describes the reaction: $\text{O}_2(\text{g}) + 2\text{SO}_2(\text{g}) \rightarrow 2\text{SO}_3(\text{g})$. What chemical does the green line refers to?

A. $\text{O}_2(\text{g})$

B. $\text{SO}_2(\text{g})$

C. $\text{SO}_3(\text{g})$



Your Turn!



For the following reaction: $\text{C}_3\text{H}_8 (\text{g}) + 5 \text{O}_2 (\text{g}) \rightarrow 3 \text{CO}_2 (\text{g}) + 4 \text{H}_2\text{O} (\text{g})$

- a) Write the expressions for the Rate of Reaction with respect to $\text{O}_2 (\text{g})$ and Rate of Reaction with respect to $\text{CO}_2 (\text{g})$.

$$r = -\frac{1}{5} \frac{\Delta[\text{O}_2]}{\Delta t} = \frac{1}{3} \frac{\Delta[\text{CO}_2]}{\Delta t}$$

- b) If the rate of change in the concentration of CO_2 ($\frac{\Delta[\text{CO}_2]}{\Delta t}$) is initially measured to be 6.0 M/s, what would be the rate of change in the concentration of O_2 ($\frac{\Delta[\text{O}_2]}{\Delta t}$) ?

$$-\frac{1}{5} \frac{\Delta[\text{O}_2]}{\Delta t} = \frac{1}{3} \times 6.0 \frac{\text{M}}{\text{s}} = 2.0 \frac{\text{M}}{\text{s}}$$

$$\frac{\Delta[\text{O}_2]}{\Delta t} = -10 \frac{\text{M}}{\text{s}}$$

Your Turn!

For the following reaction: $\text{C}_3\text{H}_8 (\text{g}) + 5 \text{O}_2 (\text{g}) \rightarrow 3 \text{CO}_2 (\text{g}) + 4 \text{H}_2\text{O} (\text{g})$

- a) Write the expressions for the Rate of Reaction with respect to $\text{O}_2 (\text{g})$ and Rate of Reaction with respect to $\text{CO}_2 (\text{g})$.

$$r = - \frac{1}{5} \frac{\Delta[\text{O}_2]}{\Delta t}$$

$$r = \frac{1}{3} \frac{\Delta[\text{CO}_2]}{\Delta t}$$

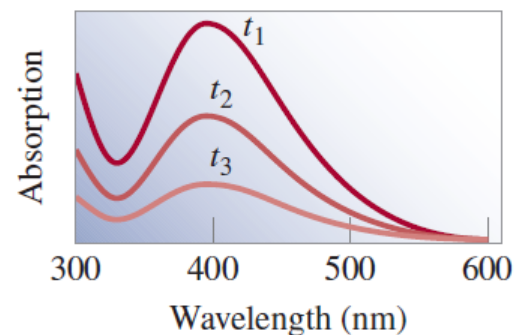
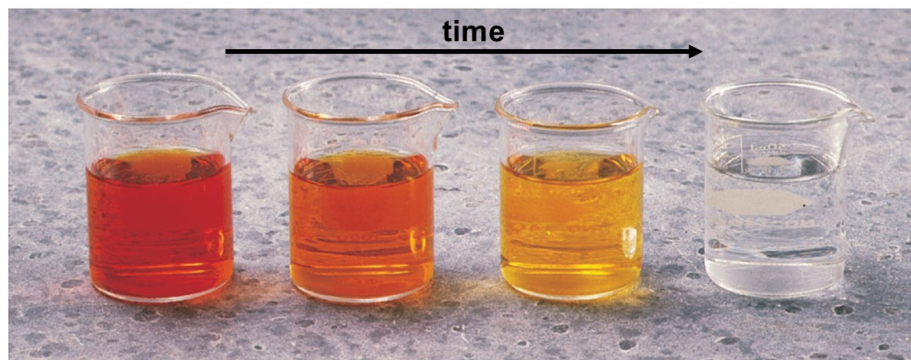
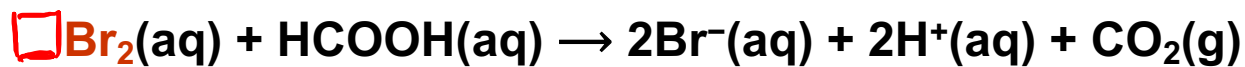
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$$- \frac{1}{5} \frac{\Delta[\text{O}_2]}{\Delta t} = \frac{1}{3} \times 6.0 \frac{\text{M}}{\text{s}}$$

$$\frac{\Delta[\text{O}_2]}{\Delta t} = -10. \frac{\text{M}}{\text{s}}$$

Measuring Average Reaction Rate

- r = the change in concentration of a **reactant**/**product** over time (Δt)



- Average reaction rate**

$$0\text{s to } 100\text{ s: } r = \frac{1}{1} \frac{-\Delta[\text{Br}_2]}{\Delta t} = \frac{-(0.00846\text{M} - 0.0120\text{M})}{100\text{ s} - 0\text{ s}} = 3.54 \times 10^{-4} \frac{\text{M}}{\text{s}}$$

$$100\text{s to } 200\text{ s: } r = \frac{1}{1} \frac{-\Delta[\text{Br}_2]}{\Delta t} = \frac{-(0.00596\text{M} - 0.00846\text{M})}{200\text{ s} - 100\text{ s}} = 2.50 \times 10^{-4} \frac{\text{M}}{\text{s}}$$

Time (s)	[Br ₂] (M)
0.0	0.0120
50.0	0.0101
100.0	0.00846
150.0	0.00710
200.0	0.00596
250.0	0.00500

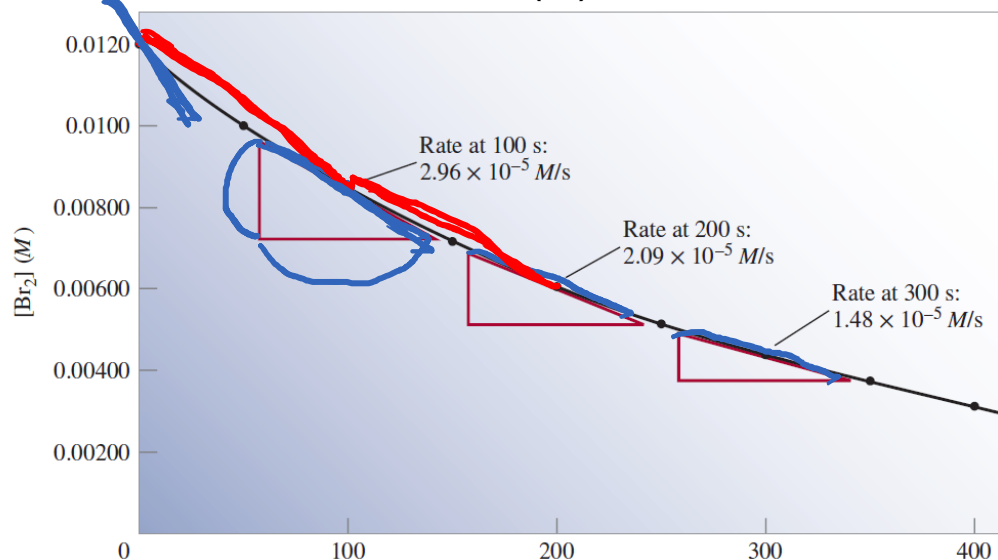
Instantaneous Reaction Rate



- Instantaneous** reaction rate: slope of tangent

$$r = -\frac{\Delta[\text{Br}_2]}{\Delta t} = -\frac{[\text{Br}_2]_{\text{final}} - [\text{Br}_2]_{\text{initial}}}{t_{\text{final}} - t_{\text{initial}}} \xrightarrow{\Delta t \rightarrow 0} -\frac{d[\text{Br}_2]}{dt}$$

- Initial reaction rate (r_0): the instantaneous reaction rate at 0 s.



t (s)	r_{average} (M/s)	$r_{\text{Instantaneous}}$ (M/s)
100		2.96×10^{-5}
100-200	2.50×10^{-5}	
200		2.09×10^{-5}

- The average rate is between the instantaneous rates at just before the interval and at just after the interval.
- The reaction rate becomes slower over time as the reactant concentration decreases.

Your Turn!

Consider reaction: $2 \text{HI (g)} \rightarrow \text{H}_2 \text{(g)} + \text{I}_2 \text{(g)}$ for the following question. The concentration of HI vs time at 508°C is given on the left.

- 1) Calculate the average rate of reaction during the time interval between 0 s to 100 s.

$$r = -\frac{1}{2} \times \frac{0.058 \text{ M} - 0.10 \text{ M}}{100 \text{ s} - 0 \text{ s}} = 2.1 \times 10^{-4} \frac{\text{M}}{\text{s}}$$

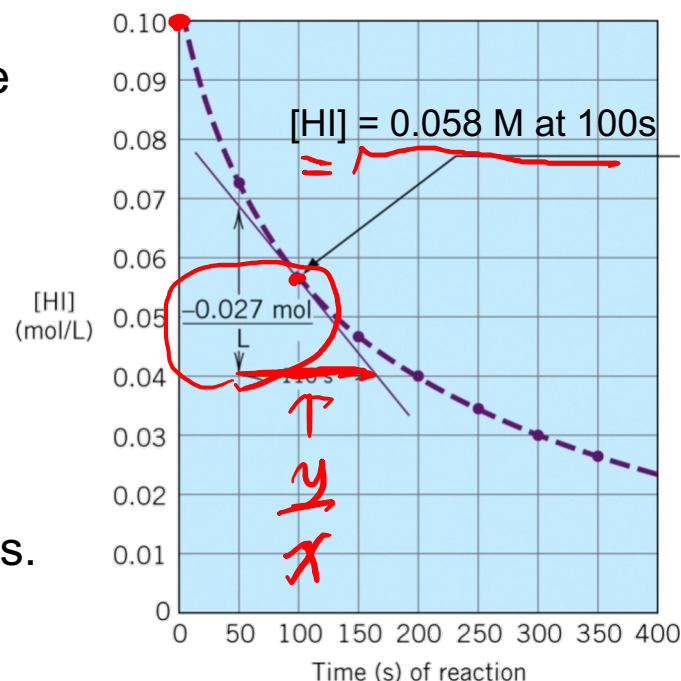
- 2) Determine the rate of change in the concentration of H_2 ($\frac{\Delta[\text{H}_2]}{\Delta t}$) during the time interval between 0 s to 100 s.

$$r = \frac{1}{1} \frac{\Delta[\text{H}_2]}{\Delta t} = 2.1 \times 10^{-4} \frac{\text{M}}{\text{s}}$$

- 3) Calculate the instantaneous rate of reaction at 100 s.

$$r = -\frac{1}{2} \times \left(\frac{d[\text{HI}]}{dt} \right) = -\frac{1}{2} \times \frac{-0.027 \frac{\text{mol}}{\text{L}}}{100 \text{ s}} = 1.2 \times 10^{-4} \frac{\text{M}}{\text{s}}$$

↑ slope



1) $2.1 \times 10^{-4} \text{ M/s}$; 2) $2.1 \times 10^{-4} \text{ M/s}$; 3) $1.2 \times 10^{-4} \text{ M/s}$

Your Turn!

Consider reaction: $2 \text{HI (g)} \rightarrow \text{H}_2 \text{(g)} + \text{I}_2 \text{(g)}$ for the following question. The concentration of HI vs time at at 508 °C is given on the left.

- 1) Calculate the average rate of reaction during the time interval between 0 s to 100 s.

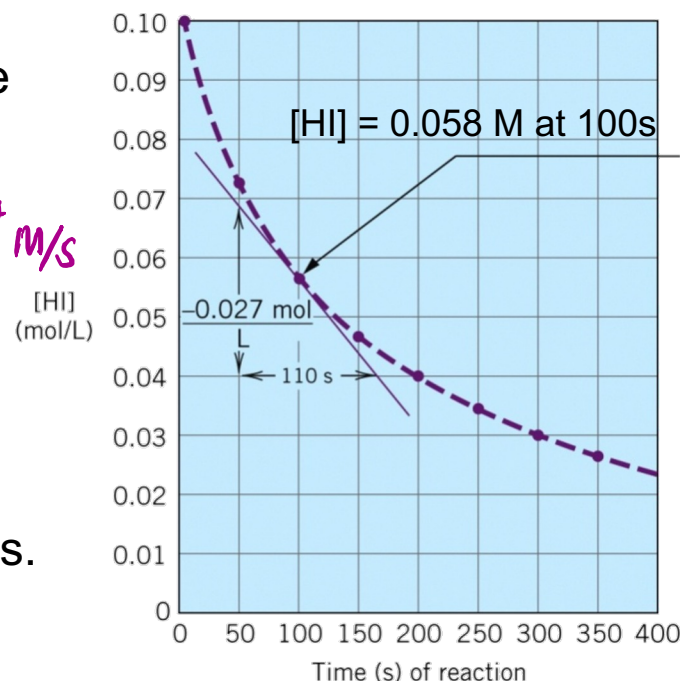
$$r = -\frac{1}{2} \frac{\Delta[\text{HI}]}{\Delta t} = -\frac{1}{2} \times \frac{0.058 \text{ M} - 0.10 \text{ M}}{100 \text{ s} - 0 \text{ s}} = 2.1 \times 10^{-4} \text{ M/s}$$

- 2) Determine the rate of change in the concentration of H_2 ($\frac{\Delta[\text{H}_2]}{\Delta t}$) during the time interval between 0 s to 100 s.

$$r = \frac{1}{1} \frac{\Delta[\text{H}_2]}{\Delta t} = 2.1 \times 10^{-4} \text{ M/s}$$

- 3) Calculate the instantaneous rate of reaction at 100 s.

$$r = -\frac{1}{2} \frac{d[\text{HI}]}{dt} = -\frac{1}{2} \times \frac{-0.027 \text{ M}}{110 \text{ s}} = 1.2 \times 10^{-4} \text{ M/s}$$



Announcement 4/1

Assignments that are due soon

- Pre-Class Assignment Week 9-F

Chem RePS Session

Wed 2:00 - 3:30 PM at KC 230

1st Chem RePS or SI worksheets for Unit 3 will be closed on 11:59 PM 4/8

Office Hour

- Dr. Wang's: Thu 10-11 AM on Zoom

SI Session

Saige Douglass – Wed 5:30 – 7:30 PM
at Hashinger 412

Paige Kwan - Thu 5 - 7 PM at KC134



The Rate Law



- The reaction rate slow down over time as the Br_2 decreases.

- Two factors influence the reaction rate:

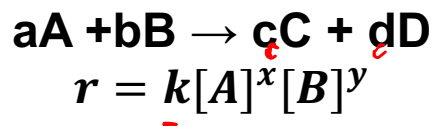
- Concentration of reactants

[reactant] $\uparrow$, r $\uparrow$

- Temperature

T $\uparrow$, r $\uparrow$

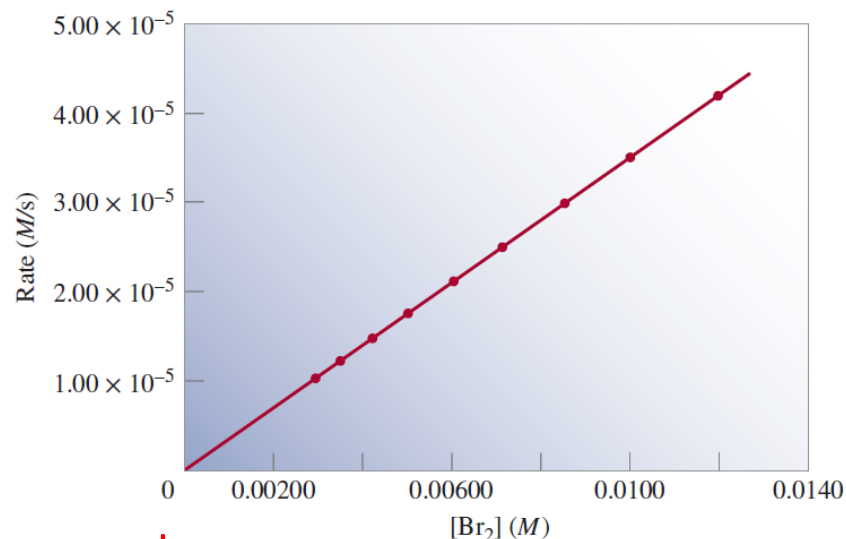
- The Rate Law:



- k : the rate constant ($\propto$ T)

- x and y : the reaction order ($x + y$) (NOT the coefficient in the equation!)

- Unit of k depends on the order of the reaction.



$\downarrow$

Time (s)	Rate (M/s)	$k = \frac{\text{rate}}{[\text{Br}_2]} (\text{s}^{-1})$
0.0	4.20×10^{-5}	3.50×10^{-3}
50.0	3.52×10^{-5}	3.49×10^{-3}
100.0	2.96×10^{-5}	3.50×10^{-3}
150.0	2.49×10^{-5}	3.51×10^{-3}
200.0	2.09×10^{-5}	3.51×10^{-3}
250.0	1.75×10^{-5}	3.50×10^{-3}

The Rate Law $r = k[A]^x[B]^y$

Zero order reaction ($x + y = 0$):

- $r = k$ (unit of $k = \frac{M}{s}$) $\frac{M}{s} = k$

The rate is independant to the concentration

First order reaction ($x + y = 1$): $\frac{M}{s} = k \times M$

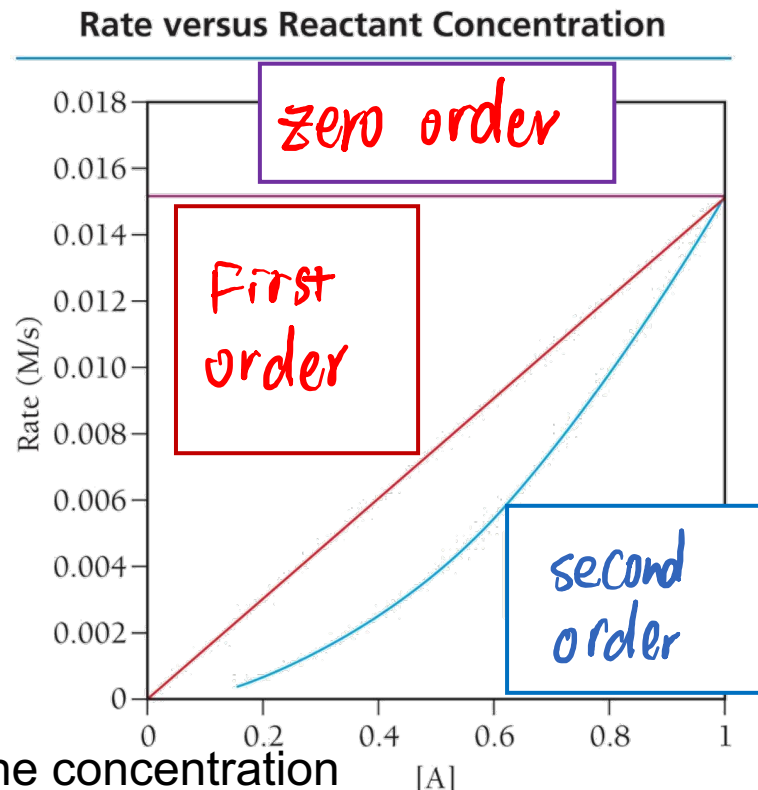
- $r = k[A]$ (unit of $k = \frac{1}{s}$)

The rate is linear to the concentration

Second order reaction ($x + y = 2$): $\frac{M}{M \cdot s} = k \times M^2$

- $r = k[A]^2$ (unit of $k = \frac{1}{M \cdot s}$)

The rate is directly proportional to square of to the concentration



For a particular reaction in which $A \rightarrow \text{Products}$, a doubling of the concentration of A causes the reaction rate to double. What is the order of the reaction? $\times 2$

A. 0 $r = 1$

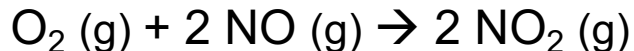
B. 1 $r = 2$

C. 2 $r = 4$



Your Turn!

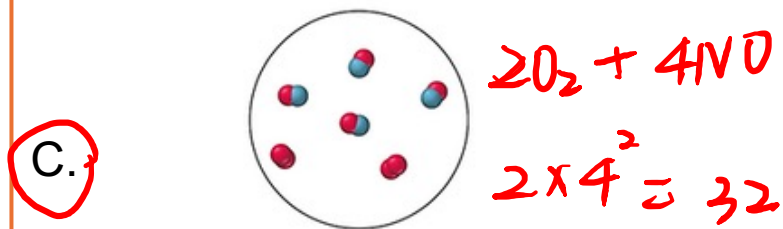
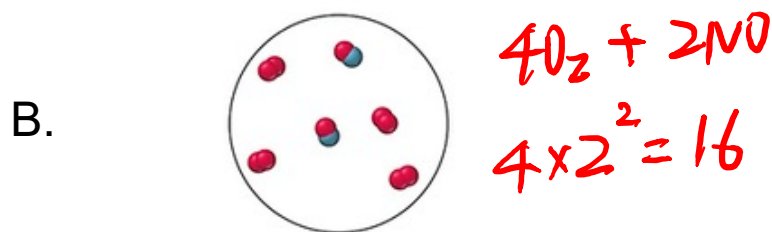
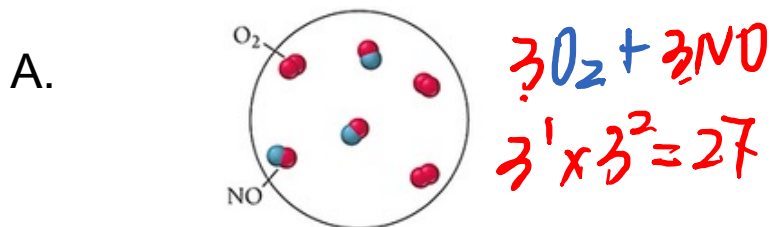
Consider the reaction below:



Knowing that the reaction is first order with respect to $\text{O}_2(\text{g})$ and is second Order with respect to $\text{NO}(\text{g})$.

1) Write Rate Law for this reaction. $r = k [\text{O}_2]^1 [\text{NO}]^2$ order = 3

2) Which mixture has the fastest initial rate?



3) If the concentration of NO was decreased by half and the concentration of O_2 was increased by four, by what factor would the reaction rate increase?

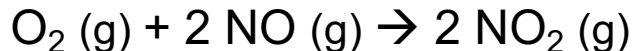
- A. 0.5 C. 2 E. 8
☒ B. 1 D. 4 F. 16

$$\left(\frac{1}{2}\right)^2 \cdot (4)^1 = \frac{1}{4} \cdot 4 = 1$$



Your Turn!

Consider the reaction below:

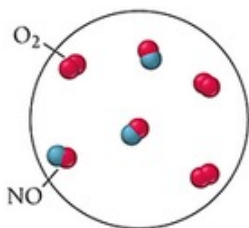


Knowing that the reaction is first order with respect to $\text{O}_2(\text{g})$ and is second Order with respect to $\text{NO}(\text{g})$.

1) Write Rate Law for this reaction. $r = k[\text{O}_2]^1[\text{NO}]^2$

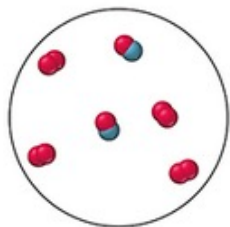
2) Which mixture has the fastest initial rate?

A.



$$\begin{aligned} [\text{O}_2] &= 3 \\ [\text{NO}] &= 3 \\ r &= k \times 27 \end{aligned}$$

B.



$$\begin{aligned} [\text{O}_2] &= 4 \\ [\text{NO}] &= 2 \\ r &= k \times 16 \\ [\text{O}_2] &= 2 \end{aligned}$$

C.



$$\begin{aligned} [\text{NO}] &= 4 \\ r &= k \times 32 \end{aligned}$$

3) If the concentration of NO was decreased by half and the concentration of O_2 was increased by four, by what factor would the reaction rate increase?

A. 0.5

C. 2

E. 8

☒ B. 1

D. 4

F. 16

$$\begin{aligned} r &= k \times (4[\text{O}_2])^1 \times \left(\frac{1}{2}[\text{NO}]\right)^2 \\ &= k \times 4 \times \frac{1}{4} \times [\text{O}_2][\text{NO}] \\ &= k[\text{O}_2][\text{NO}] \end{aligned}$$



Your Turn!

2HI (g) $\rightarrow$ H₂ (g) + I₂ (g) behaves with 2nd order kinetics. The rate constant is 0.08029 M⁻¹s⁻¹.

a. Write the complete Rate Law

$$r = 0.08029 \frac{1}{M \cdot s} [HI]^2$$

b. If the initial concentration of [HI] is 0.358M, what is the rate of reaction?

$$r = 0.08029 \frac{1}{M \cdot s} \times (0.358M)^2 = 0.0103 \frac{M}{s}$$

c.
d. Will rate INCREASE or DECREASE if [HI] = 0.100M

Your Turn!

$2\text{HI (g)} \rightarrow \text{H}_2\text{(g)} + \text{I}_2\text{(g)}$ behaves with 2nd order kinetics. The rate constant is $0.08029 \text{ M}^{-1}\text{s}^{-1}$.

a. Write the complete Rate Law

$$r = k[\text{HI}]^2$$

b. If the initial concentration of $[\text{HI}]$ is 0.358M , what is the rate of reaction?

$$r = 0.08029 \text{ M}^{-1}\cdot\text{s}^{-1} \times (0.358\text{M})^2$$

$$= 0.0103 \text{ M/s}$$

d. Will rate INCREASE or DECREASE if $[\text{HI}] = 0.100\text{M}$

Determine the Rate Law from Experiments

[A] (M)	Initial Rate (M/s)
① 0.10	0.015
0.20	0.030
③ 0.40	0.060

$A \rightarrow B$
 $r = k[A]^x$

$$\frac{r_3}{r_1} = \frac{0.060}{0.015} = \frac{k[A]_3^x}{k[A]_1^x} = \left(\frac{0.40}{0.10}\right)^x$$

$$\frac{4}{1} = 4^x \quad \Rightarrow \quad x = \log_4 4 = 1$$

What is the order of the reaction (i) and reaction (ii)? The experimental results are given below.

(i)

[A] (M)	Initial Rate (M/s)
0.10	0.015
0.20	0.015
0.40	0.015

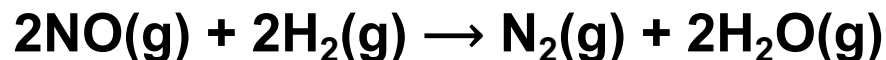
(ii)

[A] (M)	Initial Rate (M/s)
0.10	0.015
0.20	0.060
0.40	0.240

- A. (i) is 0, (ii) is 1 C. (i) is 1, (ii) is 0
 B. (i) is 0, (ii) is 2 D. (i) is 2, (ii) is 0
 E. (i) is 1, (ii) is 2 E. (i) is 2, (ii) is 1



Determine the Rate Law from Experiments



Experiment	[NO] (M)	[H ₂] (M)	Initial Rate (M/s)
1	5.0×10^{-3}	2.0×10^{-3}	1.3×10^{-5}
2	10.0×10^{-3}	2.0×10^{-3}	5.0×10^{-5}
3	10.0×10^{-3}	4.0×10^{-3}	10.0×10^{-5}

$$r = k[\text{NO}]^x[\text{H}_2]^y$$

1) $x = ?$ $y = ?$

To determine x , find the data that [H₂] is not changed: 1 & 2

$$\frac{r_2}{r_1} = \frac{5}{1.3} = 3.8 = \left(\frac{10}{5}\right)^x = 2^x \quad \begin{matrix} 3.8 = 2^x \\ x = \log_2 3.8 \approx 2 \end{matrix}$$

To determine y , find the data that [NO] is not changed: 2 & 3

$$\frac{r_3}{r_2} = \frac{10}{5} = \left(\frac{4}{2}\right)^y \quad \begin{matrix} 2 = 2^y \\ y = 1 \end{matrix}$$

The order of the reaction is: $r = k[\text{NO}]^2[\text{H}_2]^1$

order = 3

Announcement 4/3

Assignments that are due soon

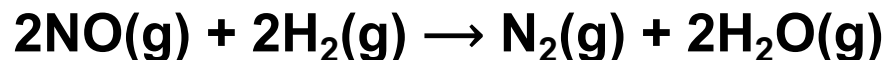
- Pre-Class Assignment Week 10-M
exam 2 wrapper (11:59 PM 4/6)

Office Hour

- Dr. Wang's: Fri 1-2 PM at KC 382



Determine the Rate Law from Experiments



Experiment	[NO] (M)	[H ₂] (M)	Initial Rate (M/s)
1	5.0×10^{-3}	2.0×10^{-3}	1.3×10^{-5}
2	10.0×10^{-3}	2.0×10^{-3}	5.0×10^{-5}
3	10.0×10^{-3}	4.0×10^{-3}	10.0×10^{-5}

$$r = k[\text{NO}]^x[\text{H}_2]^y$$

1) $x = ?$ $y = ?$

To determine x , find the data that [H₂] is not changed: ① & ②

$$\frac{r_2}{r_1} = \frac{k \times (10.0 \times 10^{-3})^x \times (2.0 \times 10^{-3})^y}{k \times (5.0 \times 10^{-3})^x \times (2.0 \times 10^{-3})^y} = \frac{5.0 \times 10^{-5} \text{ M/s}}{1.3 \times 10^{-5} \text{ M/s}}$$

$$\left(\frac{2}{1}\right)^x = 3.8$$

To determine y , find the data that [NO] is not changed: ② & ③ $x \approx 2$

$$\frac{r_3}{r_2} = \frac{k \times (10.0 \times 10^{-3})^x \times (4.0 \times 10^{-3})^y}{k \times (10.0 \times 10^{-3})^x \times (2.0 \times 10^{-3})^y} = \frac{10.0 \times 10^{-5} \text{ M/s}}{5.0 \times 10^{-5} \text{ M/s}}$$

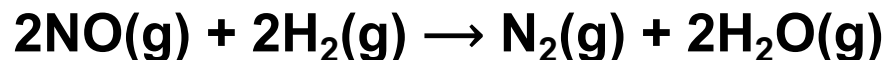
$$\left(\frac{2}{1}\right)^y = 2$$

$$y = 1$$

The order of the reaction is:

$$r = k[\text{NO}]^2[\text{H}_2]^1$$

Determine the Rate Law from Experiments



Experiment	[NO] (M)	[H ₂] (M)	Initial Rate (M/s)
1	5.0×10^{-3}	2.0×10^{-3}	1.3×10^{-5}
2	10.0×10^{-3}	2.0×10^{-3}	5.0×10^{-5}
3	10.0×10^{-3}	4.0×10^{-3}	10.0×10^{-5}

$$r = k[\text{NO}]^2[\text{H}_2]^1$$

2) $k = ?$

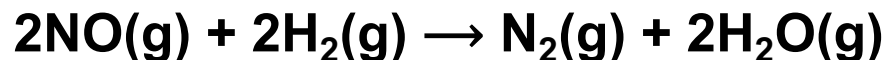
$$k = \frac{r}{[\text{NO}]^2[\text{H}_2]} = \frac{5.0 \times 10^{-5} \frac{\text{M}}{\text{s}}}{(10.0 \times 10^{-3} \text{ M})^2 \times (2.0 \times 10^{-3} \text{ M})} = 2.5 \times 10^{-2} \frac{1}{\text{s} \cdot \text{M}^2} = 250 \text{ s}^{-1} \cdot \text{M}^{-2}$$

3) when $[\text{NO}] = 12.0 \times 10^{-3} \text{ M}$ and $[\text{H}_2] = 6.0 \times 10^{-3} \text{ M}$, $r = ?$

$$r = 250 \text{ s}^{-1} \cdot \text{M}^{-2} \times (12.0 \times 10^{-3} \text{ M})^2 \times (6.0 \times 10^{-3} \text{ M})$$

$$= 2.2 \times 10^{-4} \frac{\text{M}}{\text{s}}$$

Determine the Rate Law from Experiments



Experiment	[NO] (M)	[H ₂] (M)	Initial Rate (M/s)
1	5.0×10^{-3}	2.0×10^{-3}	1.3×10^{-5}
2	10.0×10^{-3}	2.0×10^{-3}	5.0×10^{-5}
3	10.0×10^{-3}	4.0×10^{-3}	10.0×10^{-5}

$$r = k[\text{NO}]^2[\text{H}_2]^1$$

2) $k = ?$

$$r_2 = k \times (10.0 \times 10^{-3} \text{ M})^2 \times (2.0 \times 10^{-3} \text{ M}) = 5.0 \times 10^{-5} \text{ M/s}$$

$$k = 250 \text{ M}^{-2} \cdot \text{s}^{-1}$$

3) when $[\text{NO}] = 12.0 \times 10^{-3} \text{ M}$ and $[\text{H}_2] = 6.0 \times 10^{-3} \text{ M}$, $r = ?$

$$r = 250 \text{ M}^{-2} \cdot \text{s}^{-1} \times (12.0 \times 10^{-3} \text{ M})^2 \times (6.0 \times 10^{-3} \text{ M}) = 2.2 \times 10^{-4} \text{ M/s}$$

Your Turn!

Consider the reaction between nitrogen dioxide and carbon monoxide:



The initial rate of the reaction is measured at several different concentrations of the reactants, and the tabulated results are shown here.

From the data, determine:

(a) the rate law for the reaction.

Handwritten calculations for the rate law:

$[\text{NO}_2]^x$: ① & ② ② & ④ ① & ④

$$\left(\frac{0.4}{0.1}\right)^x = \frac{0.033}{0.0021} \approx 15.7 \quad 4^x = 16$$

$$x = \frac{\log 15.7}{\log 4} = 2$$

$[\text{CO}]^y$: ② & ③ ② & ④

$$\left(\frac{0.2}{0.1}\right)^y = 1 \quad 2^y = 1 \quad y = 0 = \frac{\log 1}{\log 2} = 0$$

	$[\text{NO}_2] (\text{M})$	$[\text{CO}] (\text{M})$	Initial Rate (M/s)
①	0.10	0.10	0.0021
②	0.20	0.10	0.0082
③	0.20	0.20	0.0083
④	0.40	0.10	0.033

(b) the rate constant (k) for the reaction.

Handwritten rate law: $r = k [\text{NO}_2]^2$

Your Turn!

Consider the reaction between nitrogen dioxide and carbon monoxide:



The initial rate of the reaction is measured at several different concentrations of the reactants, and the tabulated results are shown here.

From the data, determine:

(a) the rate law for the reaction.

$$r = k[\text{NO}_2]^x[\text{CO}]^y$$

$$x(\textcircled{1} \& \textcircled{2}): \left(\frac{0.20}{0.10}\right)^x = \frac{0.0082}{0.0021} \approx 4$$

$$2^x = 4 \quad x = 2$$

$$y(\textcircled{2} \& \textcircled{3}): \left(\frac{0.20}{0.10}\right)^y = \frac{0.0083}{0.0082} \approx 1$$

$$2^y = 1 \quad y = 0$$

	[NO ₂](M)	[CO](M)	Initial Rate (M/s)
①	0.10	0.10	0.0021
②	0.20	0.10	0.0082
③	0.20	0.20	0.0083
④	0.40	0.10	0.033

$$r = k[\text{NO}_2]^2$$

(b) the rate constant (k) for the reaction.

$$r_1 = k[0.10\text{M}]^2 \approx 0.0021\text{ M/s}$$

$$k \approx 0.21\text{ M}^{-1}\text{s}^{-1}$$

Your Turn!

Consider the reaction below:



Given the data on the right, determine rate constant k (with units) of the rate law for this reaction.

$[\text{CHCl}_3] (\text{M})$	$[\text{Cl}_2] (\text{M})$	Initial Rate (M/s)
0.010	0.010	0.0035
0.020	0.010	0.0069
0.020	0.020	0.0098
0.040	0.040	0.027

Your Turn!

Consider the reaction below:



Given the data on the right, determine rate constant k (with units) of the rate law for this reaction.

$[\text{CHCl}_3] (\text{M})$	$[\text{Cl}_2] (\text{M})$	Initial Rate (M/s)
0.010	0.010	0.0035
0.020	0.010	0.0069
0.020	0.020	0.0098
0.040	0.040	0.027

$$r = k [\text{CHCl}_3]^x [\text{Cl}_2]^y$$

$x: \textcircled{1} \& \textcircled{2}$
 $2^x = 2$
 $x = 1$

$$y: \textcircled{2} \& \textcircled{3}$$

$$\left(\frac{0.020}{0.010} \right)^x = \frac{0.0069}{0.0035}$$

$$2^y = 1.420$$

$$y = \log_2 1.420 = 0.5$$

$$r_1 = k [0.010 \text{ M}] [0.010 \text{ M}]^{\frac{1}{2}} = 0.0035 \frac{\text{M}}{\text{s}}$$

$$k = 3.5 \text{ M}^{-\frac{1}{2}} \cdot \text{s}^{-1}$$